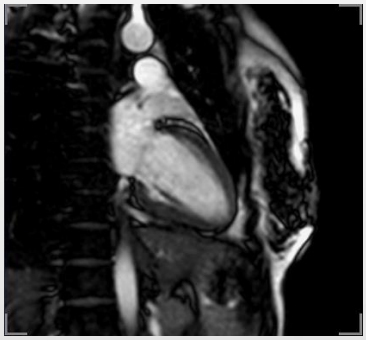
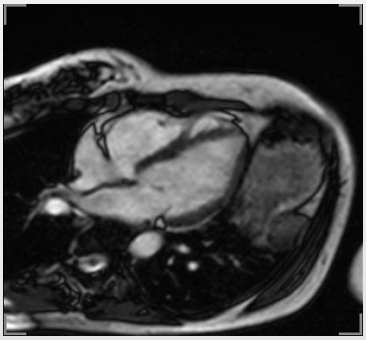
＊“mCardiac package” and “EasyTech Cardiac” are required.

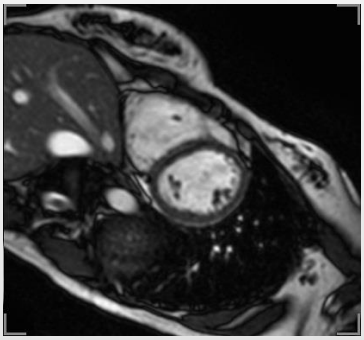
Cardiac MR : Planning（CardioLine）

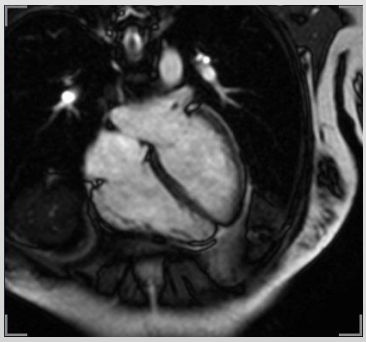
Introduction

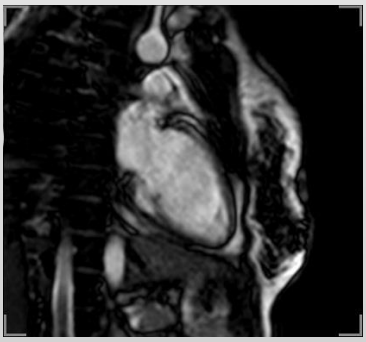
Cardiac MRI has the following basic cross sections.
・LV two chamber(Vertical long)
・Horizontal long
・LV short Axial
・Four chamber
・Three chamber
This section describes how to set up each cross section using CardioLine (cardiac positioning support application).
This application can be used to detect the 6 standard views of the heart.
-Displays and confirms the detected cross section and can be adjusted accordingly.
・ Set the imaging positioning ROI in this section.

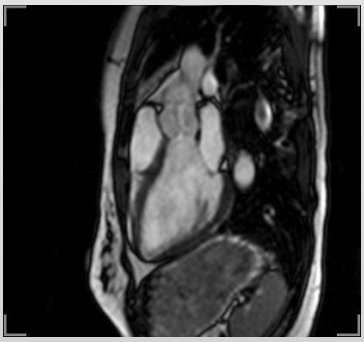

Vertical long


Horizontal long


LV short Axial


Four chamber

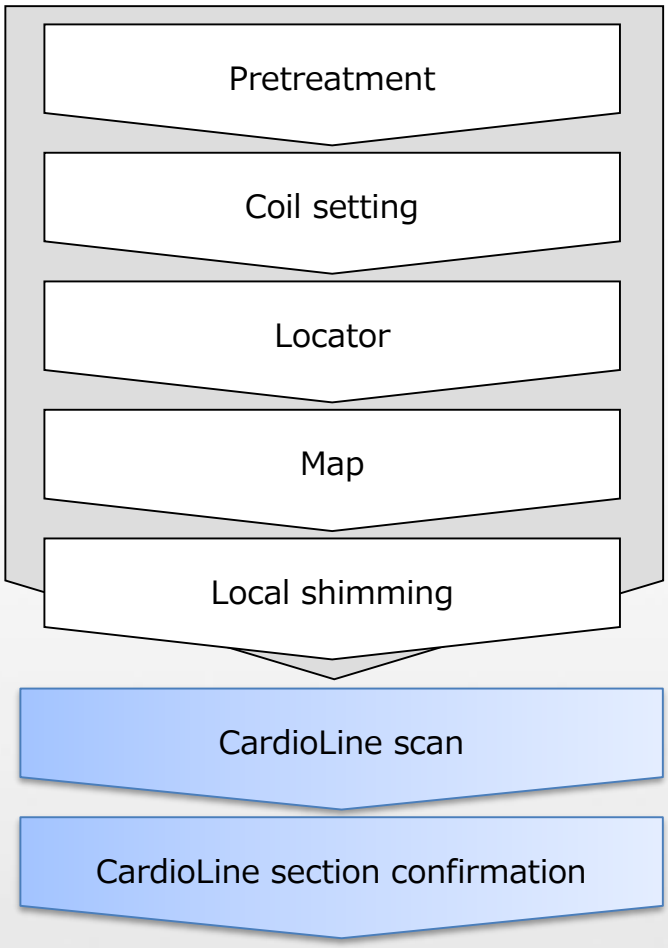

LV two chamber


Three chamber

Imaging Conditions

・All images are taken with ECG/PPG and also breath-hold.
・Use the CardioLine Imaging Conditions without changing the settings.

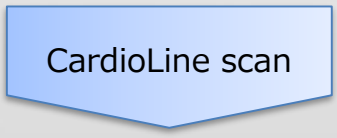
● scanning workflow



Planning trans axial images that covers the whole heart.

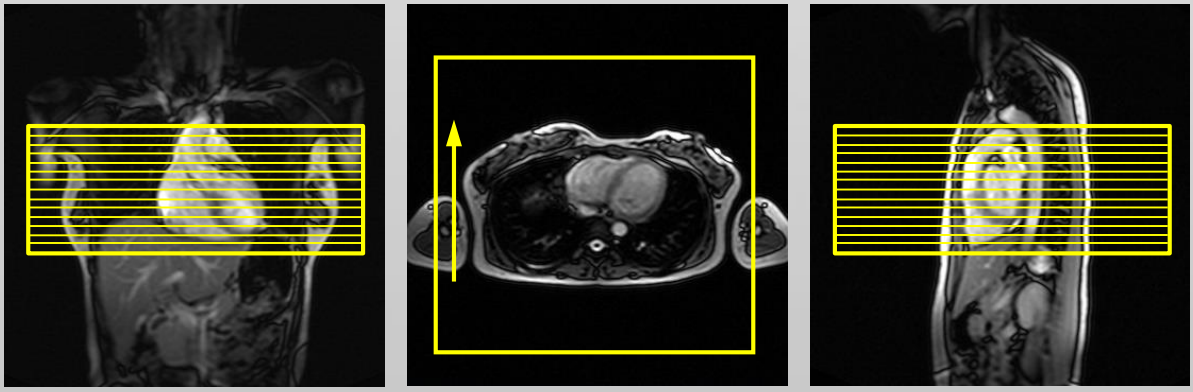
Check the detected cross section.

● Scanning Workflow



Trans axial image covering the whole heart is taken.

(1) Adjust the slices and FOV to cover the whole heart. Slices coverage should above the Aortic Outflow(Aortic Valve) to below the Apex of the heart as a guide.



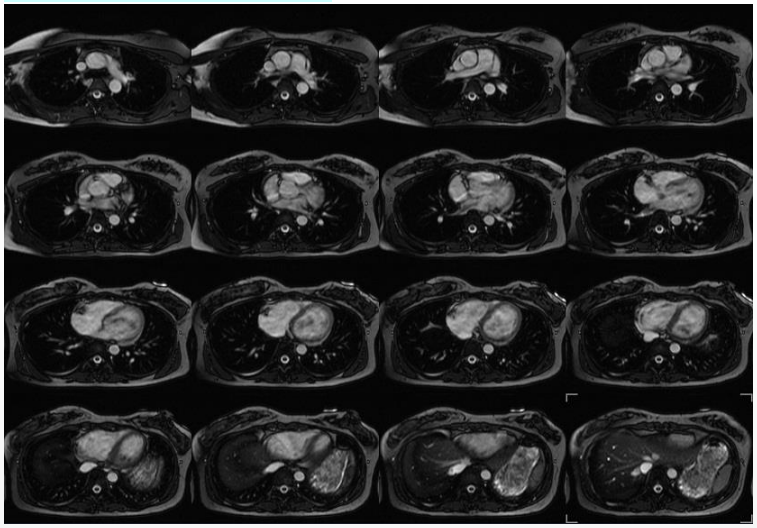
(2) Set the gate condition.

・Read the patient's R-R interval / heart rate.

・Set the delay time to the maximum so that the actual collection is in diastole.

（3） Take an image with a breath-hold.

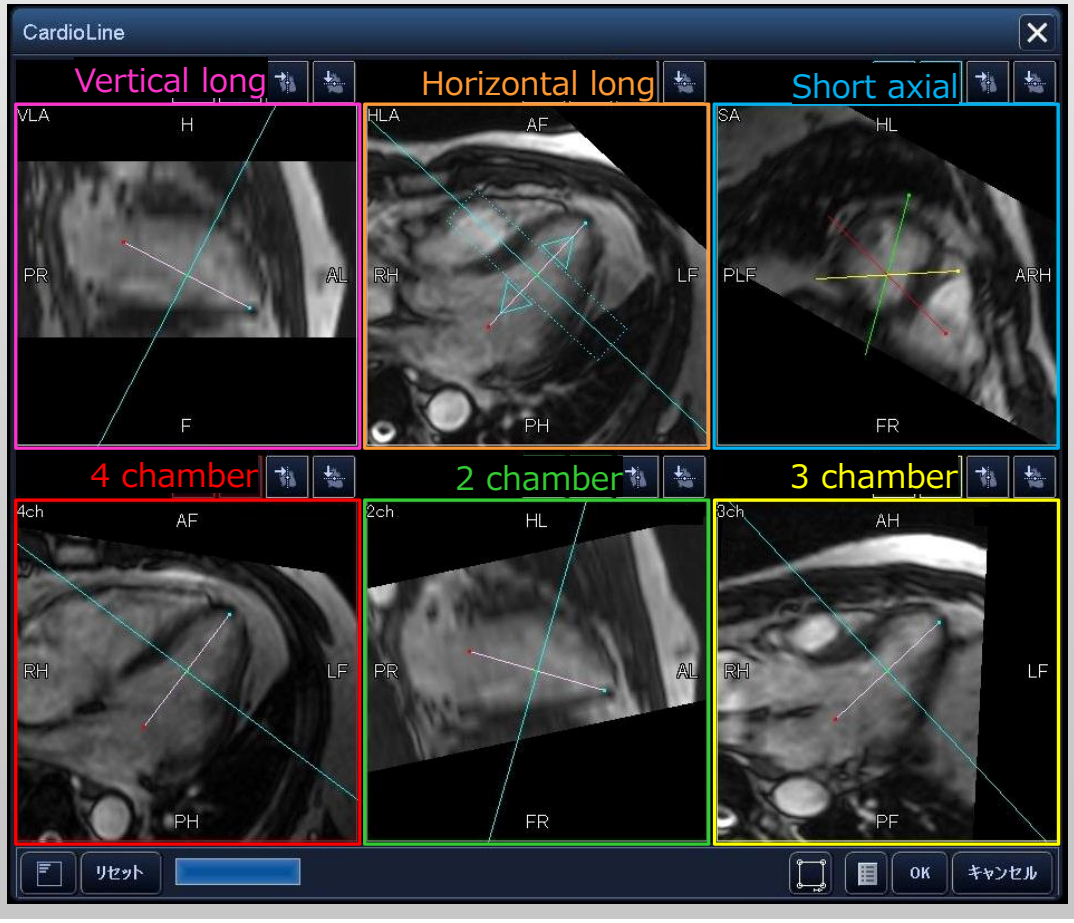
CardioLine images



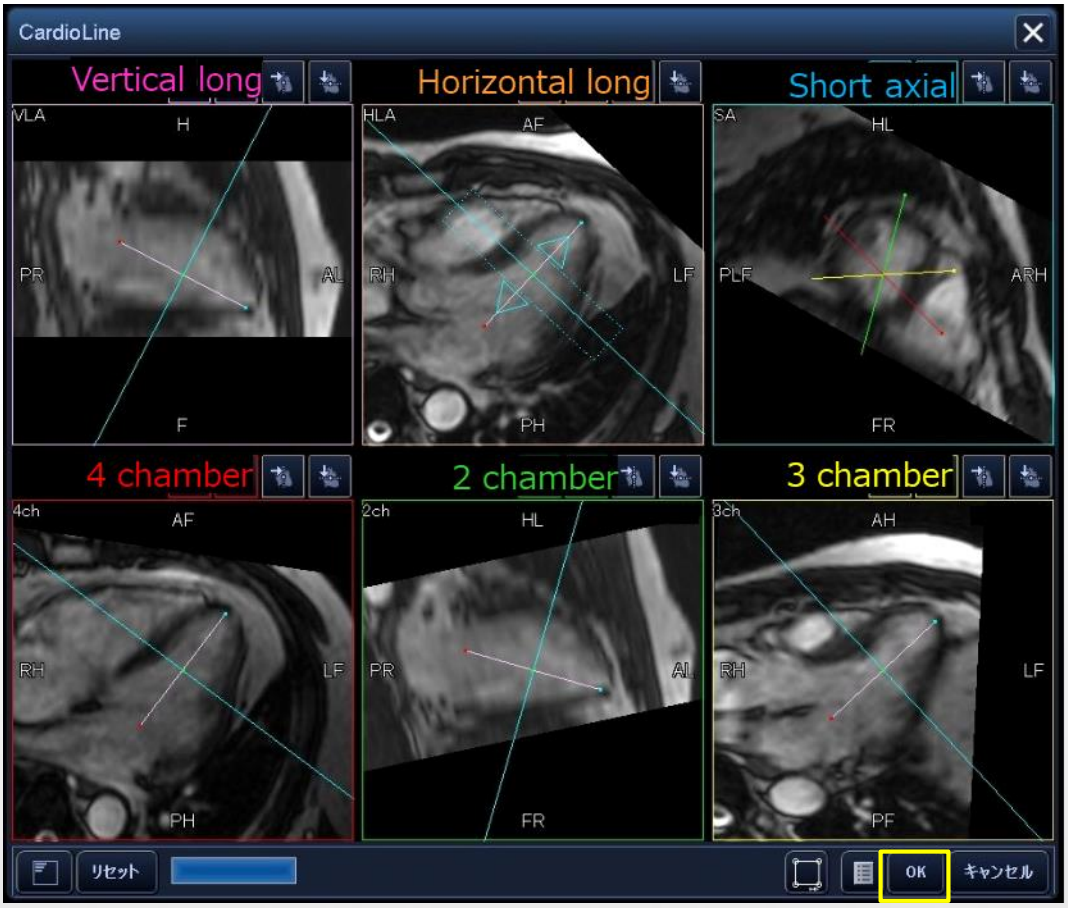
CardioLine section confirmation

Check the imaging section set by Cardioline

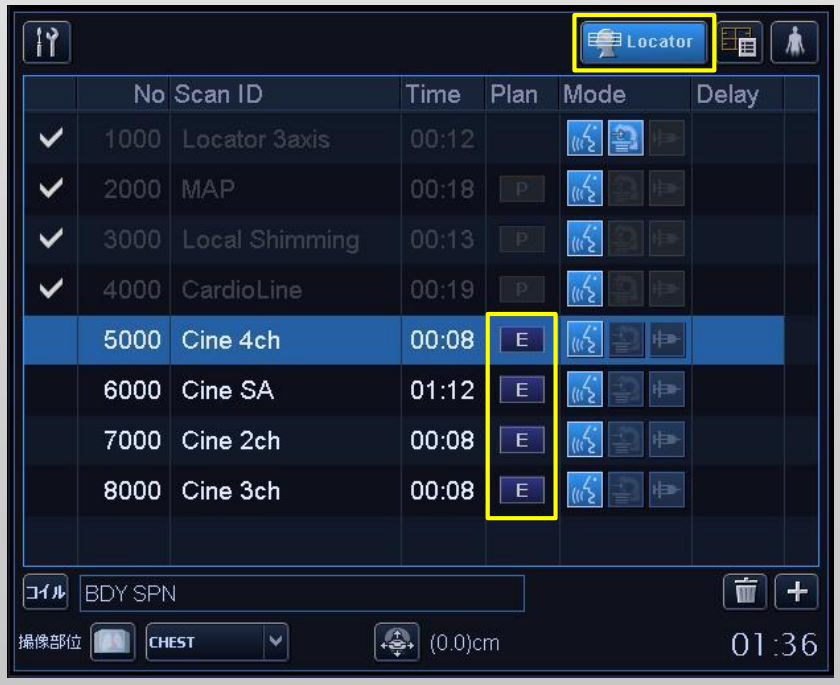
（1） When the imaging is completed, the CardioLine window will open and the detection cross section will be displayed.
Check the slice cross section and fine-tune the angle here.
The color of ROI on the image corresponds to the color of the frame of each cross section.



(2) When you are satisfied with the cross section, click [OK]. The adjusted cross-section information and image are saved and the CardioLine window closes.



(3) If the Plan has an E mark in the sequence cue list, it is preset to reflect the cross section set by CardioLine for that sequence.
Click [Locator], check the ROI in the scan plan, click [Que & Exit], and then start imaging.



Provided by "Canon Medical Systems MRI application support service"